

Lab 13 – Code Refactoring Using AI Assistance

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Task 1: Refactoring – Removing Code Duplication Prompt

Refactor the following legacy Python code by removing duplicated logic and creating a reusable function to calculate the area and perimeter of rectangles. Add proper docstrings.

Legacy Code

```
Assignment13.py
1  print("Area of Rectangle:", 5 * 10)
2  print("Perimeter of Rectangle:", 2 * (5 + 10))
3  print("Area of Rectangle:", 7 * 12)
4  print("Perimeter of Rectangle:", 2 * (7 + 12))
5  print("Area of Rectangle:", 10 * 15)
6  print("Perimeter of Rectangle:", 2 * (10 + 15))
```

OUTPUT :

```
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
```

Refactored Code

```
def rectangle_details(length, width):
```

```
    """
```

```
    Calculates and prints the area and perimeter of a rectangle.
```

```
    Parameters:
```

```
    length (int or float): Length of the rectangle
```

```
    width (int or float): Width of the rectangle
```

```
    """    area = length *
```

```
    width    perimeter = 2
```

```
    * (length + width)
```

```
    print("Area of
```

```
    Rectangle:", area)
```

```
    print("Perimeter of
```

```
Rectangle:", perimeter)
rectangle_details(5, 10)
rectangle_details(7, 12)
rectangle_details(10,
15)
```

```
Assignment13.py > rectangle_details
1  def rectangle_details(length, width):
2      """
3      💡 Calculates and prints the area and perimeter of a rectangle.
4      Parameters:
5      length (int or float): Length of the rectangle
6      width (int or float): Width of the rectangle
7      """
8      area = length * width
9      perimeter = 2 * (length + width)
10     print("Area of Rectangle:", area)
11     print("Perimeter of Rectangle:", perimeter)
12     rectangle_details(5, 10)
13     rectangle_details(7, 12)
14     rectangle_details(10, 15)
```

OUTPUT :

```
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
```

Explanation

- Repeated code blocks were identified.
- A reusable function `rectangle_details()` was created.
- This improves readability and maintainability.
- If calculations change later, only the function needs modification.

Task 2: Refactoring – Optimizing Loops and Conditionals Prompt

Optimize the following code by replacing inefficient nested loops with a faster data structure such as a set.

Legacy Code

```
Assignment13.py > ...
1  names = ["Alice", "Bob", "Charlie", "David"]
2  search_names = ["Charlie", "Eve", "Bob"]
3
4  for s in search_names:
5      found = False
6      for n in names:
7          if s == n:
8              found = True
9
10     if found:
11         print(f"{s} is in the list")
12     else:
13         print(f"{s} is not in the list")
```

OUTPUT :

```
Charlie is in the list
Eve is not in the list
Bob is in the list
```

Refactored Code

```
names = ["Alice", "Bob", "Charlie", "David"]
search_names = ["Charlie", "Eve", "Bob"]
name_set = set(names)
for name in search_names:
    if name in name_set:
        print(f'{name} is in the list')
    else:
        print(f'{name} is not in the list')
```

```
Assignment13.py > ...
2  search_names = ["Charlie", "Eve", "Bob"]
3
4  name_set = set(names)
5
6  for name in search_names:
7      if name in name_set:
8          print(f'{name} is in the list')
9      else:
10         print(f'{name} is not in the list')
```

OUTPUT :

```
Charlie is in the list
Eve is not in the list
Bob is in the list
```

Explanation

- Nested loops have **$O(n^2)$ complexity**.
- Using a **set** provides **$O(1)$ lookup time**.
- This improves performance significantly for large datasets.

Performance Comparison

Method	Time Complexity
Nested Loop	$O(n \times m)$
Set Lookup	$O(n + m)$

Task 3: Refactoring – Extracting Reusable Functions Prompt

Refactor the following script by extracting price and tax calculations into a reusable function called `calculate_total(price)`.

Legacy Code

```
Assignment13.py > ...
4  print("Total Price:", total)
5
6  price = 500
7  tax = price * 0.18
8  total = price + tax
9  print("Total Price:", total)
```

OUTPUT :

```
Total Price: 295.0
Total Price: 590.0
```

Refactored Code

```
def calculate_total(price):
```

```
    """
```

```
    Calculates total price including tax.
```

```
    Parameters:    price (float): Base
price of the product
```

Returns:

float: Total price including tax

"""

tax = price * 0.18

total = price + tax

return total

```
print("Total Price:", calculate_total(250)) print("Total  
Price:", calculate_total(500))
```

```
Assignment13.py > ...  
1  def calculate_total(price):  
2      """  
3      Calculates total price including tax.  
4  
5      Parameters:  
6      price (float): Base price of the product  
7  
8      Returns:  
9      float: Total price including tax  
10     """  
11     tax = price * 0.18  
12     total = price + tax  
13     return total  
14  
15       
16     print("Total Price:", calculate_total(250))  
17     print("Total Price:", calculate_total(500))
```

OUTPUT :

```
Total Price: 295.0  
Total Price: 590.0
```

Explanation

- Repeated logic moved into **calculate_total()**.
- Improves **reusability** and **clean code structure**.
- Function can be reused across different modules.

Task 4: Refactoring – Replacing Hardcoded Values with Constants Prompt

Identify magic numbers in the code and replace them with named constants.

Legacy Code

```
Assignment13.py
1 print("Area of Circle:", 3.14159 * (7 ** 2))
2 print("Circumference of Circle:", 2 * 3.14159 * 7)
```

OUTPUT :

```
Area of Circle: 153.93791
Circumference of Circle: 43.98226
```

Refactored Code $PI = 3.14159$ $RADIUS = 7$

area = $PI * (RADIUS ** 2)$ circumference = $2 *$

$PI * RADIUS$ print("Area of Circle:", area)

print("Circumference of Circle:", circumference)

```
Assignment11.3.py U Assignment13.py U X
Assignment13.py > ...
1 PI = 3.14159
2 RADIUS = 7
3
4 area = PI * (RADIUS ** 2)
5 circumference = 2 * PI * RADIUS
6
7 print("Area of Circle:", area)
8 print("Circumference of Circle:", circumference)
```

OUTPUT :

```
Area of Circle: 153.93791
Circumference of Circle: 43.98226
```

Explanation

- Hardcoded values are called **magic numbers**.
- Replacing them with **constants** improves readability.
- Constants allow **easy updates in one place**.

Task 5: Refactoring – Improving Variable Naming and Readability Prompt

Improve the readability of the following code by using descriptive variable names and comments.

Legacy Code

```
Assignment11.3.py U Assignment13.py U X
Assignment13.py > ...
1 a = 10
2 b = 20
3 c = a * b / 2
4 print(c)
```

OUTPUT:

```
100.0
```

Refactored Code

```
# Base and height of a triangle
base = 10 height = 20
# Formula for area of triangle triangle_area
= (base * height) / 2 print(triangle_area)
```

```
Assignment13.py > ...
1 # Base and height of a triangle
2 base = 10
3 height = 20
4
5 # Formula for area of triangle
6 triangle_area = (base * height) / 2
7
8 print(triangle_area)
```

OUTPUT :

```
100.0
```

Explanation

- Variables **a**, **b**, **c** were unclear.
- Renamed them to **base**, **height**, **triangle_area**.
- Added comments explaining the logic.